

Flora Xuan

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Technical Skills

Programming Languages: Python, TypeScript, C++, C, SQL, HTML, VHDL, MATLAB, Assembly Language

Frameworks & Tools: FastAPI, Jenkins (CI/CD), React, PyTorch, REST APIs, MySQL, Git, Linux, Postman, VS Code

Cloud Services: AWS Bedrock, Lambda, API Gateway

AI / Machine Learning: RAG, NLP, Prompt Engineering, LLM Evaluation, Computer Vision, CNNs, AWS Bedrock Knowledge Bases

Education

University of Toronto

St. George Campus, Toronto
Expected January 2027

Bachelor of Applied Science in Computer Engineering

Minor in Artificial Intelligence Engineering

Academic Achievements: Edward S Rogers Sr. Admission Scholarship \$1,000 | Dean's Merit Award \$2,000

Relevant Coursework: Deep Learning, Artificial Intelligence, Databases, Computer Networks, Algorithms & Data Structures, Operating Systems

Experience

Multimedia Drivers Team Co-op

Markham, Canada

ATI Technologies ULC ("AMD")

May 2023 – Apr 2024

- Served as tech lead for the VQ Automation tool, selected as a top-50 project for AMD Canada's 2024 Innovation Showcase
- Maintained and upgraded the multimedia-driver test farm, set up a new MySQL database, and optimized plot server algorithms
- Supported multimedia-driver validation by debugging and evaluating video-quality for internal engineering requests
- Performed quality evaluation tasks upon teammate requests

Projects

Dementia Care Support Chatbot (RAG-based AI System)

University of Toronto

RAG Service & Evaluation | Python, FastAPI, AWS Bedrock, RAG Pipeline

Sep 2025 – Apr 2026

- Developed a **RAG-based** AI chatbot delivering grounded, citation-backed dementia caregiving guidance with actionable responses
- Enabled reliable multi-turn interactions with session-based memory and consistent response formatting (4–6 key recommendations)
- Achieved system latency **under 10 seconds** through model and pipeline optimization
- Evaluated performance on 100+ caregiver queries, achieving **~83% response success** and **0.87 faithfulness score**
- Applied prompt engineering, response tuning, and safety guardrails to improve accuracy and enforce controlled outputs
- Developed with doctors' input and continuously enhanced for ongoing professional dementia-care use

Ancient Ceramic Classification (PyTorch CNN)

University of Toronto

Deep Learning Course Project | Python, PyTorch, CNNs, Computer Vision

Sep 2025 – Jan 2026

- Developed a PyTorch-based CNN to classify ancient ceramics across 10 stylistic categories
- Built an image preprocessing pipeline for 11,000+ images with cleaning, normalization, augmentation, and train/validation/test splits
- Designed a three-block CNN using BatchNorm, SiLU, MaxPooling, dropout, and Adam optimization
- Evaluated model generalization using held-out test data, confusion matrices, and baseline comparisons

Video Quality Automation (VQA) — AMD

ATI Technologies ULC ("AMD")

Technical Lead | Python, Jenkins, MySQL, Test Automation, Test-Farm Infrastructure

May 2023 – Apr 2024

- Designed and improved automation efficiency by implementing Jenkins job triggers, reducing manual execution steps
- Set up a MySQL-backed test-farm pipeline for automated job management, quality-score processing, and results visualization
- Optimized quality score evaluation by analyzing and comparing multiple calculation methods for scalability

Interactive Map – GIS Application Development

University of Toronto

Team Leader and Contact Person | C++, Linux, Git, UI implementation & Optimization

Feb 2022 – May 2022

- Led a team of three to develop an Interactive Geographic Information System application
- Implemented the map UI with Day/Night display modes
- Launched features to reduce load and run time by **~85%** including search algorithms and other optimization functions in C++